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BIOE 582 Spring 2013
Project 1b Part 1

Q1. What information can you find in the Dilution data set? Show the slot names of the Dilution data set.

```
#list slot names  
getSlots('AffyBatch')  
#get slot information  
?AffyBatch
```

cdfName: name of the CDF file the data in the AffyBatch came from

nrow: number of rows in the data array

ncol: number of columns in the data array

assayData: matrix of raw data or standard errors if parameter 'sd' was set to TRUE when reading the CDF file

phenoData: phenotype information for sample

annotation: string identifying the annotation to be used in an ExpressionSet, where it identifies the platform on which the sample was assayed.

protocolData: protocol information for the sample

featureData: feature-level (e.g., probeset-level) information.

experimentData: containing experiment information.

.__classVersion__: the R and Biobase version number used to create the instance.

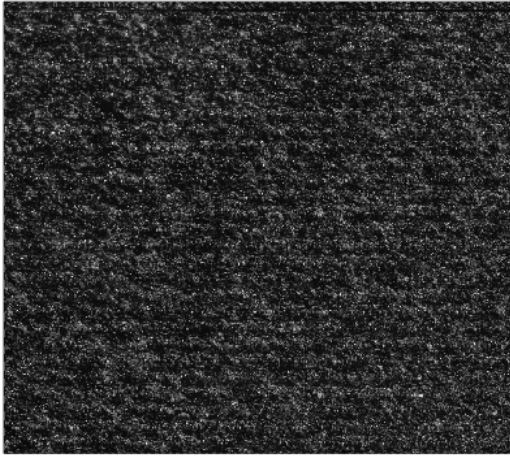
Q2. How do you obtain the phenotypic information from the Dilution data set?

```
#get the phenotype data for each sample  
pData(Dilution@phenoData)  
#get the labels for the data listed in pData  
varLabels(Dilution@phenoData)  
#get the description of the labels seen in varLabels/pData  
varMetadata(Dilution@phenoData)
```

Q3. For array 20B in the Dilution data set, generates spatial image of log base 2 probe intensities.

```
image(Dilution[,2],transfo=log)
```

20B



Q4: Generate histogram, boxplot for all the arrays. Make sure your curves and boxes have different colors for 20 and 10. What information can you obtain from the plots?

#histogram

hist(Dilution, col=c("red","red","blue","blue"))

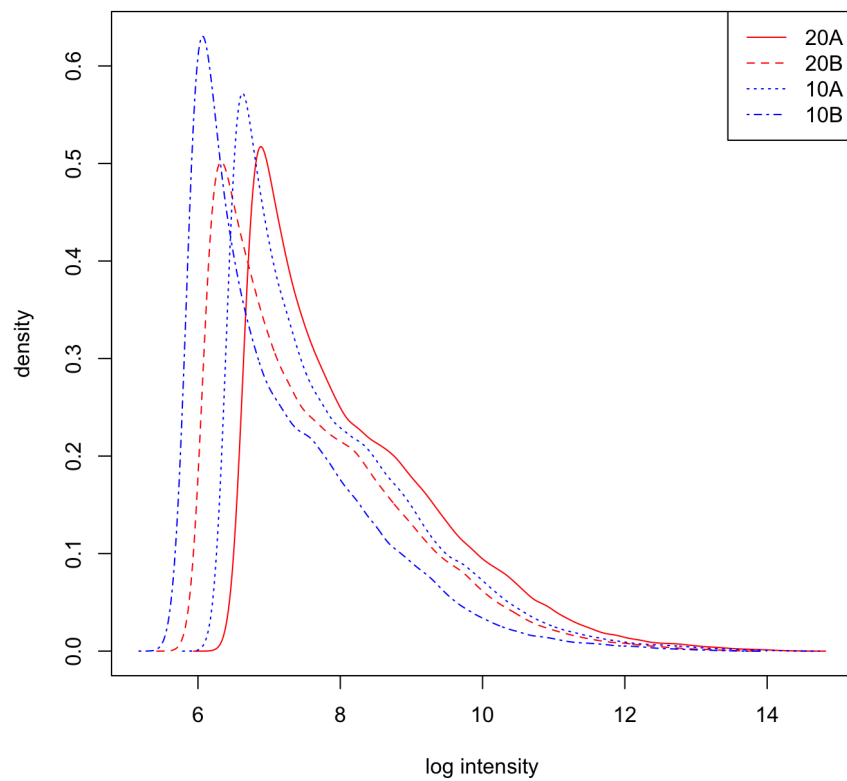
legend("topright",sampleNames(Dilution),col=c("red","red","blue","blue"),lty=c(1,2,3,4))

#boxplot

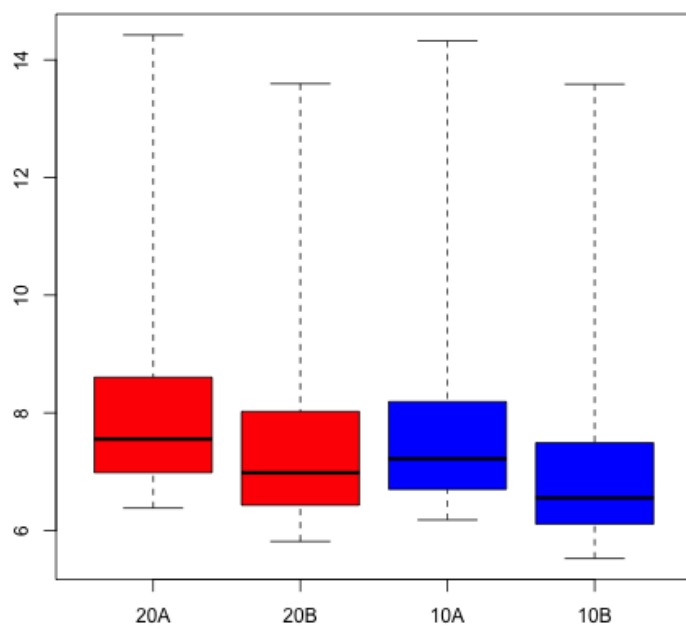
*boxplot(Dilution, col=c("red","red","blue","blue"), names=sampleNames(Dilution),
main="Pre-Normalization")*

Because Affymetrix microarrays contain only 1 sample per microarray, normalization across arrays is very important. Both of these plots show the differences between the samples (mean, distribution, median), clearly indicating need for normalization due to the large differences even between samples of the same phenotype, which arise due to various differences in each microarray.

Pre-Normalization Histogram



Pre-Normalization Boxplot



Q5. Perform the background adjustment for the Dilution data set using any of the background correction methods implemented in affy package.

```
dbg = bg.correct(Dilution,method="rma")
```

Q6. Show the names of normalization methods implemented in affy package. Perform the normalization using the default method.

```
normalize.methods(Dilution)
```

```
[1] "constant"      "contrasts"      "invariantset"   "loess"          "methods"
[6] "qspline"       "quantiles"      "quantiles.robust"
```

```
#defaults to quantile normalization
```

```
dn = normalize(dbg)
```

Q7. Generate histogram, boxplot and MA scatter plots for all the normalized arrays. Use different colors as in Q4.

```
#histogram
```

```
hist(dn, col=c("red","red","blue","blue"),main="Post-Normalization Histogram")
```

```
legend("topright",sampleNames(Dilution),col=c("red","red","blue","blue"),lty=c(1,2,3,4))
```

```
#boxplot
```

```
boxplot(dn, col=c("red","red","blue","blue"), names=sampleNames(Dilution),
```

```
main="Post-Normalization Boxplot")
```

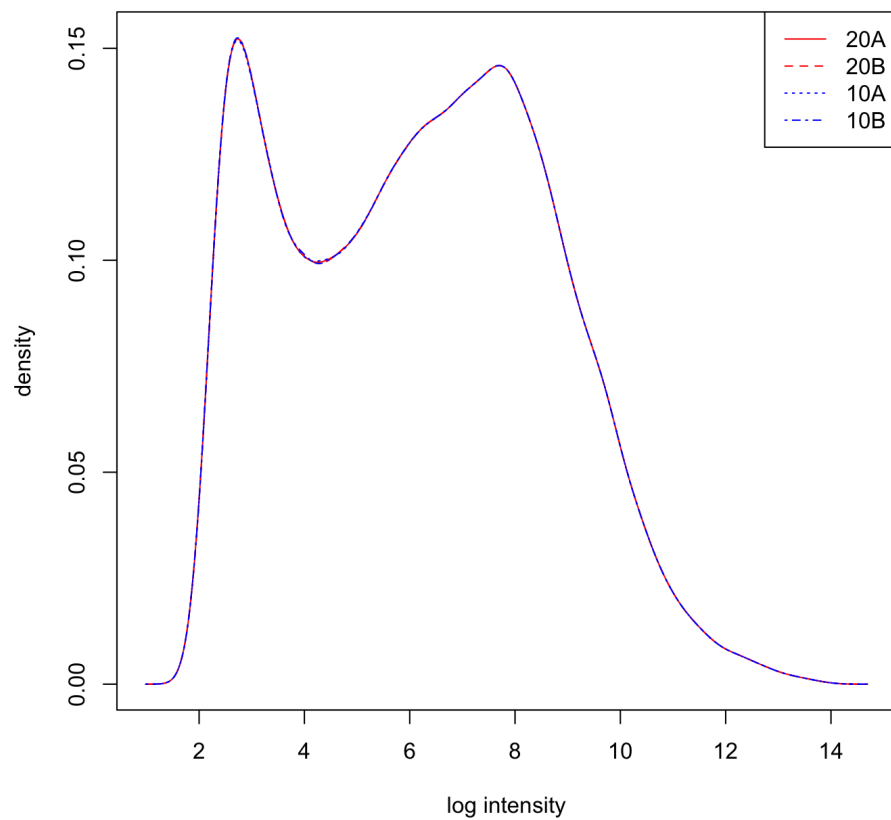
```
#MAplot
```

```
dev.new(width=7,height=7)
```

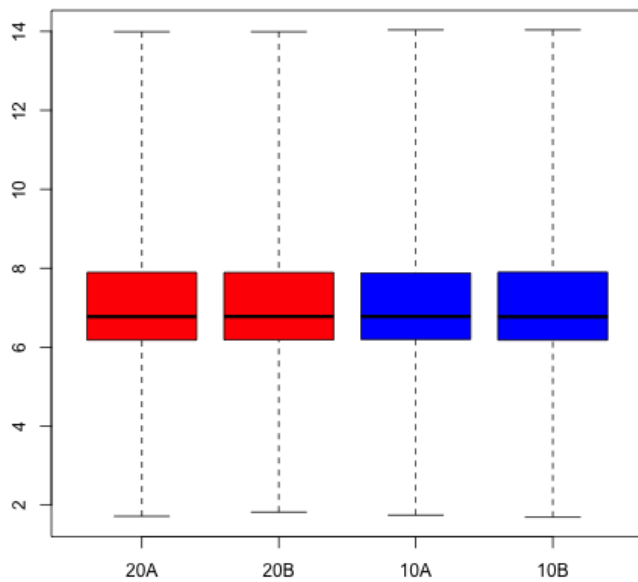
```
par(mfrow=c(2,2))
```

```
MAplot(dn)
```

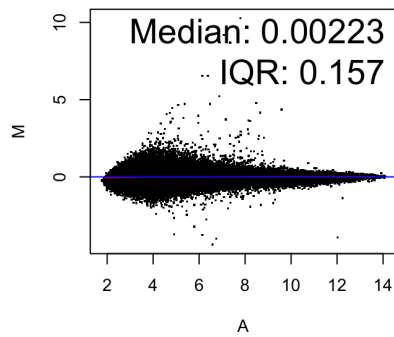
Post-Normalization Histogram



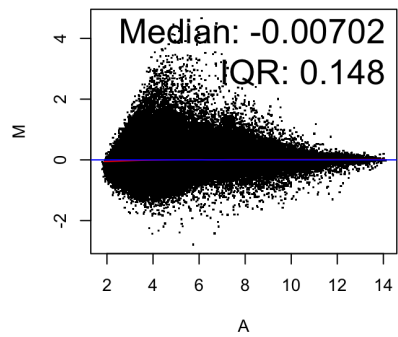
Post-Normalization Boxplot



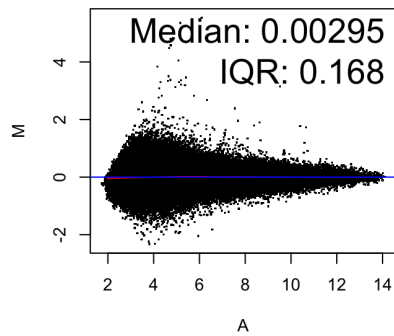
20A vs pseudo-median reference chip



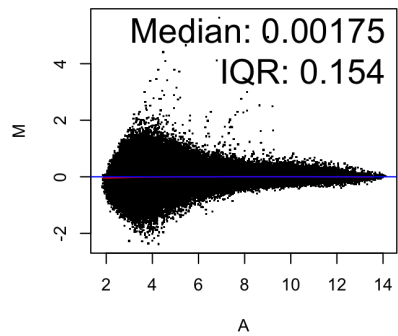
20B vs pseudo-median reference chip



10A vs pseudo-median reference chip



10B vs pseudo-median reference chip



Q8. Perform summarization of the normalized data.

```
eset=expresso(dn,bg.correct=F,normalize=F,pmcorrect.method="pmonly",summary.method="
medianpolish")
```